Webinar architecture cheat sheet

Short, code-verified answers for Agents, Webhooks, WhatsApp, Website Intake, Tools, APIs, security, and debugging.

Committed main: 8be029ce | August 2026 | No generic claims

> **One-sentence architecture** The portal saves configuration through Express into PostgreSQL; a call enters through VoBiz, Plivo, Twilio, or Webcall; `createSession()` loads the live agent and runs either classic STT → LLM/tools → TTS or Gemini Live; call end writes outcome, billing, memory, and integrations.

THE SAFEST DEMO STORY

Input
Website form, CSV, Google Sheet, Meta lead, inbound phone, API

Conversation
Sarvam/Deepgram STT, LLM, knowledge, tools, Cartesia/Sarvam TTS

Action
Booking, webhook API, email, WhatsApp, transfer, callback

Output
History, disposition, wallet usage, CRM/Sheet writeback, signed event webhook

Six architecture pictures

1 · Platform

React dashboard
↓
Express /api · auth · tenant
↓
Services · PostgreSQL · providers

2 · Voice call

VoBiz / Plivo / Twilio / Webcall
↓
Media WebSocket · createSession
↓
Classic or Gemini Live

3 · Classic voice turn

Caller audio → STT
↓
LLM + knowledge + memory
↓
Tool result → TTS

4 · Website Intake

Website JSON → token/HMAC
↓
campaign_leads + live batch
↓
Dial now or queue

5 · WhatsApp

Agent rule / send_whatsapp
↓
BSP override → Meta direct
↓
Native VoBiz fallback

6 · Tool call

User → LLM function + args
↓
Backend / external API
↓
Result → LLM → voice

Runtime truth: calls load the current `agents` row. Save changes live behavior. Publish creates a labelled snapshot. Restore writes a snapshot back to the live row and creates a draft record.

Agent settings that matter

Area	Stored	What changes at runtime
Prompt, greeting, first speaker	`agents` columns	LLM instructions and who starts
STT / LLM / TTS provider + model	`agents` columns	Adapter factories and credentials required
Voice stack	`agent_config.voiceStack`	Classic three-stage pipeline or Gemini Live
Language, endpointing, VAD, barge-in	Columns + `agent_config`	Recognition, turn end, interruption behavior
TTS voice/speed, ambient, prebuffer	Columns + `agent_config`	Voice and playback feel
Knowledge source IDs	`agent_knowledge_sources`	Per-turn RAG; vector + FTS or FTS fallback
Tools	`agents.tools` JSON	Functions visible to the LLM
Calendar capability	`agent_config`	Which booking calendar/event type is allowed
Silence, max duration, end phrases	Columns	Check-in and automatic hangup
Recording, voicemail/AMD, memory	`agent_config`	Audio storage, machine path, caller context
Post-call URL	Column + `agent_config`	One agent-specific final HTTP POST
Outcomes / after-call messages	`agent_dispositions`	Extraction, funnel status, email/WhatsApp rules
Phone assignment	`phone_numbers` + agent IDs	Inbound and outbound carrier routing

> **Two separate saves:** the Agent Save button writes agent configuration. The Outcomes tab has its own save API.

> **Auto Summarization nuance:** the visible toggle enables post-call summarization from `agent_config`. Live token-context compression reads the separate `token_summarization_enabled` column.

ALL RUNTIME TOOL TYPES

end_call · native_check_availability · native_booking · native_list_bookings · native_reschedule · native_cancel · call_transfer · transfer_cold · transfer_warm · transfer_agent · squad_followup · webhook · send_sms · send_email · send_whatsapp · cal_com

Webhooks and Website Intake

Webhook job	Actual route / behavior	Security
Website/campaign lead	`POST /api/webhooks/lead/:token` → lead + batch → dial/queue	Token; optional HMAC; Meta payload requires HMAC
WhatsApp request-a-call	`POST /api/webhooks/whatsapp/:token` → 5-min dedupe → call	Token only
Telephony callbacks	`/api/webhooks/{provider}/*` answer/status/AMD/recording	Twilio/Plivo signature; VoBiz differs
Workspace events	Customer URL receives `call.started/completed/failed`	Timestamped HMAC; 10 s; 2 retries
Agent post-call URL	One URL receives final call payload	Optional HMAC; raw fetch path
Custom HTTP tool	GET/POST mid-call; result returned to LLM	Headers/secret; SSRF guard; 8 s; 1 retry

WEBSITE SETUP

1. Create a Campaign with Website/Webhook lead source and assign an Agent.
2. Open Connections; copy `/api/webhooks/lead/{token}`.
3. Map the real payload key to phone; optionally map name and email.
4. For production, set an HMAC secret and sign the exact raw request body.
5. Use Listen mode for a sample, then POST a controlled test lead.
6. Check `campaign_leads`, queued/calling status, provider callback, call History, and outcome.

> **Duplicate risk:** generic website submissions have no phone-level dedupe. Make the source form idempotent and disable repeated submits.

> **Queue reasons:** campaign paused/stopped, outside business hours, DNC/TRAI, concurrency, attempts, wallet, or provider failure.

WhatsApp: what is really implemented

Meta direct · Implemented
WABA ID + Phone Number ID + encrypted System User token. Template fetch and mid/post-call sends.

Native VoBiz · Implemented
Own or purchased WhatsApp channel, message log, status updates, chatbot, WhatsApp calling.

BYO BSP · Implemented
SnapServe signs a call.mid/call.ended POST; Wati/Zapier/customer BSP sends the message.

Instant/shared · Partial
Existing rows work. New enrollment returns HTTP 410 and must not be promised.

OUTBOUND PRECEDENCE

Agent has BSP URL?

↓

Yes: BSP only

↓

Else Meta direct

↓

Else native VoBiz

SETUP

1. Meta path: enter Business, WABA ID, Phone Number ID, permanent token; connect to an Agent.
2. Fetch/pick an approved template; set Agent → After the call rule or add `send_whatsapp`.
3. Test with a call containing caller speech and a completed/booking outcome.
4. For request-a-call, configure the inbound token URL or native VoBiz inbound Agent.

> **Status limitation:** native VoBiz updates sent/delivered/read in `whatsapp_messages`. Meta-direct and BSP sends are not shown with the same delivery lifecycle.

> **Do not claim:** Plivo WhatsApp, direct Meta inbound chatbot/status receiver, campaign-owned WhatsApp, or new Instant WhatsApp signup.

Top 20 likely questions

#	Question	Short answer
1	How does SnapServe work?	Input → Agent session → voice/tools → outcome/integrations.
2	Does Publish make changes live?	Save does. Publish is a labelled snapshot; Restore applies it.
3	Where do I assign a number?	Phone Numbers, not Agent Save.
4	Live test vs phone call?	Same session core; browser Webcall vs real carrier.
5	What is a webhook?	An automatic HTTP request; identify its direction/job first.
6	How do I connect a website?	POST JSON to the Campaign lead token URL.
7	Can it call immediately?	Yes, unless hours/DNC/status/concurrency/wallet rules queue it.
8	How do I connect WhatsApp?	Meta direct, native VoBiz, or BYO BSP.
9	Is Instant WhatsApp available?	Not for new signup; endpoint returns 410.
10	Can it book Calendar?	Yes with scheduling tools and connected/configured calendar.
11	Can it use my API?	Yes through an HTTPS custom webhook tool.
12	How is API access authenticated?	sk_live_ API keys; dashboard uses HttpOnly JWT cookie.
13	Can I upload knowledge?	Yes; attach sources to an Agent for hybrid retrieval.
14	Does it remember callers?	Yes through caller facts/profiles when configured.
15	Which carriers?	VoBiz, Twilio, Plivo, Webcall; not Vonage/Exotel.
16	Which CRM?	HubSpot and Zoho; not Salesforce.
17	Which payment gateway?	Razorpay INR wallet; not Stripe.
18	Does it support Tamil/Hindi?	Yes, especially through Sarvam; verify selected language/model.
19	Can a caller interrupt?	Yes through configurable barge-in/VAD.
20	Where do I debug?	History → call detail/logs, then provider/webhook/DB path.

Difficult questions and demo warnings

Do not demo / promise

- Vonage, Exotel/Tata, Salesforce, Stripe, AWS Polly adapter, OpenAI Realtime.
- New Instant/shared WhatsApp enrollment.
- Cal.com or full warm-transfer briefing on Gemini Live.

Explain with caveat

- Team workspace sharing: agents/calls are workspace-aware; campaigns/KB/squads are not uniformly shared.
- Direct Meta WhatsApp: outbound works; native inbound/status/chatbot is not implemented for this path.
- Knowledge: semantic vector search can fall back to full-text only.
- Webhook delivery: retries exist, but these paths are not a durable outbound queue.
- Connections Listen: in-memory and replica/restart sensitive.

SECURITY ANSWERS

Session	Custom HS256 JWT, usually HttpOnly `vo_session`, 30-day expiry.
Developer API	`sk_live_` raw key shown once; SHA-256 hash stored.
User provider/OAuth secrets	AES-256-GCM encrypted.
Managed platform keys	Plaintext in `platform_settings` — a real security gap.
Admin	Shared `X-Admin-Token`; no per-admin RBAC in middleware.
Website intake	Token-only unless HMAC configured.
WhatsApp request-call	Token-only; no Meta signature verification on that route.
VoBiz telephony	No provider-signature equivalent; sensitive status paths use correlation/HMAC tokens.

Debugging and essential files

FIVE-STEP DEBUGGING PATTERN

1. **Name the exact path:** provider, route, Agent ID, call/lead ID, and direction.
2. **UI/request:** inspect payload, auth type, response, and client/workspace context.
3. **Database:** confirm row/column changed—not just a success toast.
4. **Runtime:** confirm `createSession` loaded the same Agent and provider keys.
5. **External callback:** inspect provider response, status webhook, retries, timeout, and final History.

Agent silent

Carrier/WS → key → STT final → LLM timeout/tool → TTS → wallet/logs.

Tool fails

Saved live tool → valid name/schema → provider support → HTTPS/SSRF → timeout.

Lead but no call

Campaign status → DNC/hours → concurrency → wallet → carrier callback.

WhatsApp missing

Provider path → BSP override → template → call gate → wallet/cap → lastError.

ESSENTIAL EVIDENCE

App/middleware	`artifacts/api-server/src/app.ts`
Routes	`artifacts/api-server/src/routes/index.ts`
Agent API / DB	`routes/agents.ts` · `lib/db/src/schema/agents.ts`
Session / voice	`orchestration/session.ts` · `pipeline.ts`
Tools / booking	`services/tool-executor.ts`
Website Intake	`routes/campaign-webhook.ts` · `campaignCaller.ts`
WhatsApp	`routes/whatsapp.ts` · `whatsapp-native.ts` · `services/whatsapp/outbound.ts`
Event subscriptions	`routes/webhook-endpoints.ts`
Dashboard Agent UI	`artifacts/dashboard/src/pages/agents/builder.tsx`

Production-only environment values, provider dashboards, and live credentials were not available for this audit. Configuration-dependent claims must be verified in the deployed environment before the webinar.